

Basic Principles of Medicine 2

Module: Endocrinology & Reproduction

DLA

Fetal Membranes

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Assigned Reading

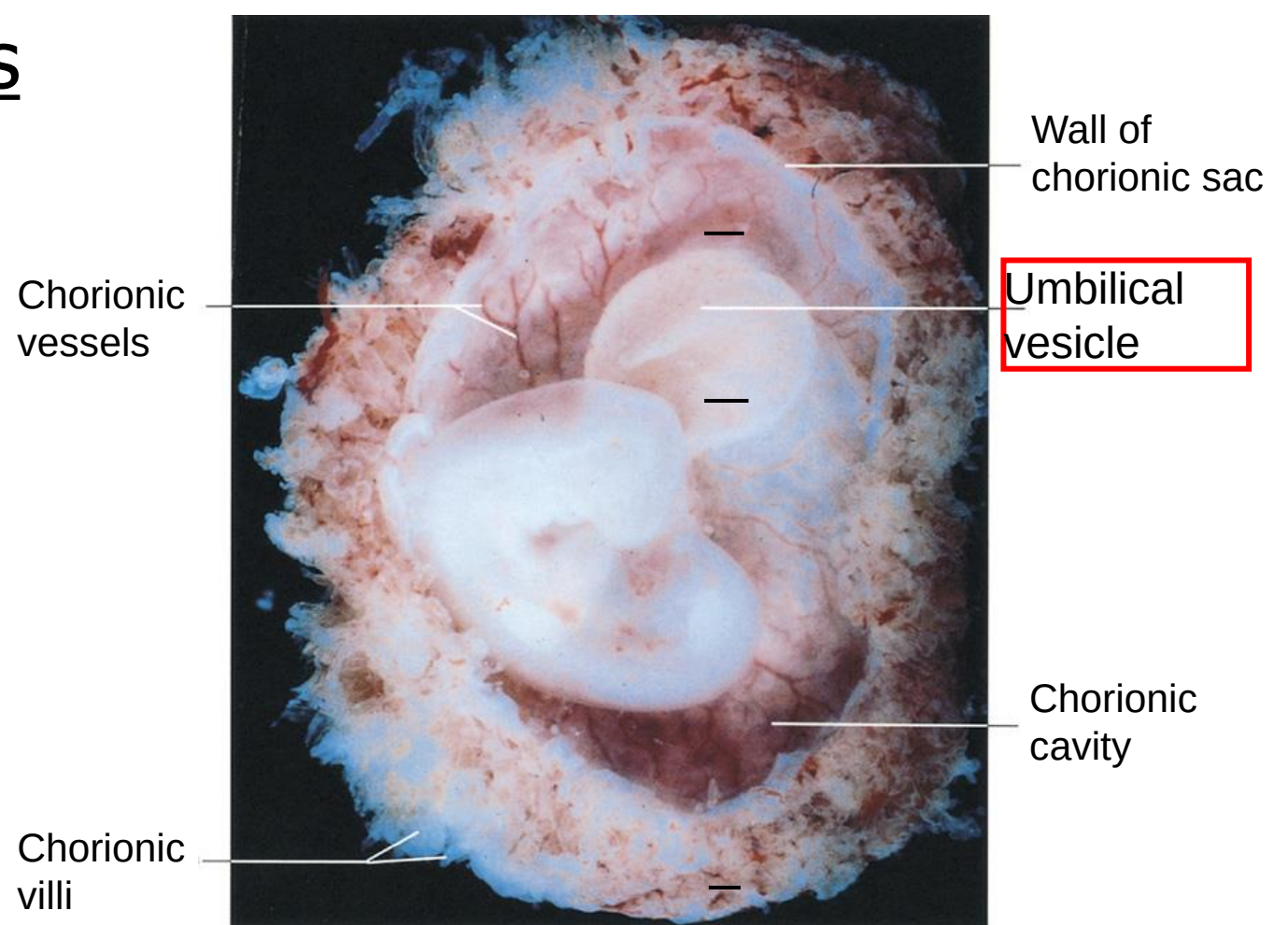
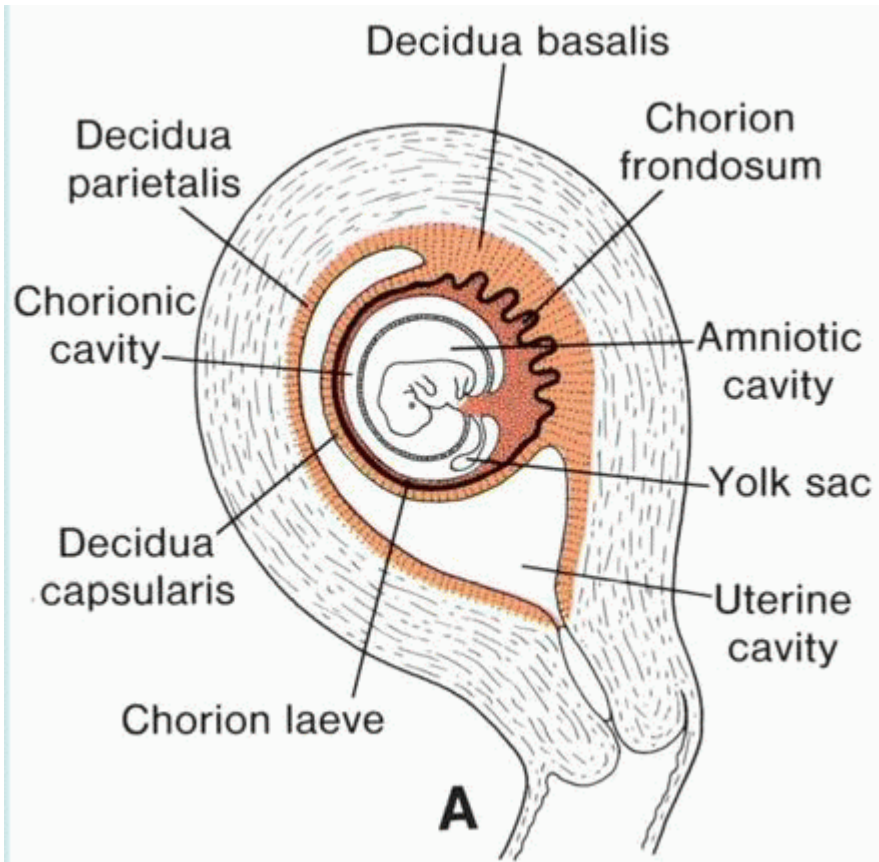
- Moore K.L, Persaud T.V.N, Torchia M.G. The Developing Human. 11th ed, Clinically Oriented Embryology Chapter 4. pg. 47-49, 59-61
- Moore K.L, Persaud T.V.N, Torchia M.G. The Developing Human. 11th ed, Clinically Oriented Embryology Chapter 7 pg. 99-128

Objectives

SOM.MKII.BPM2.1.ER.1.ANAT.1068	Describe the formation of the placenta.
SOM.MKII.BPM2.1.ER.1.ANAT.1069	Describe the fetomaternal junction.
SOM.MKII.BPM2.1.ER.1.ANAT.1070	Describe the placental circulation.
SOM.MKII.BPM2.1.ER.1.ANAT.1071	Describe how the amniotic fluid is produced and its circulation.
SOM.MKII.BPM2.1.ER.1.ANAT.1072	Describe the development of allantois and fetal membranes.
SOM.MKII.BPM2.1.ER.1.ANAT.1073	Describe placenta accreta, percreta and previa and their clinical significance
SOM.MKII.BPM2.1.ER.1.ANAT.1074	Describe hydatiform mole and its association with choriocarcinoma
SOM.MKII.BPM2.1.ER.1.ANAT.1075	Compare and contrast oligohydramnios versus polyhydramnios and describe their associated conditions
SOM.MKII.BPM2.1.ER.1.ANAT.1076	Describe the cause of hemolytic disease in a newborn (erythroblastosis fetalis)
SOM.MKII.BPM2.1.ER.1.ANAT.1077	Describe embryogenesis of monozygotic, dizygotic twins and conjoined twins
SOM.MKII.BPM2.1.ER.1.ANAT.1078	Describe the mechanism of twin to twin transfusion

Fetal Membranes

- Amnion
- Umbilical vesicle
- Allantois
- Chorion



Overview

- **Amnion**

The amnion is a membrane that closely covers the embryo when first formed.

- **Umbilical vesicle/Yolk sac (development covered in Early Embryology)**

The yolk sac is a membranous sac attached to an embryo, formed by cells of the hypoblast adjacent to the embryonic disk.

- **Allantois (development covered in CPR module)**

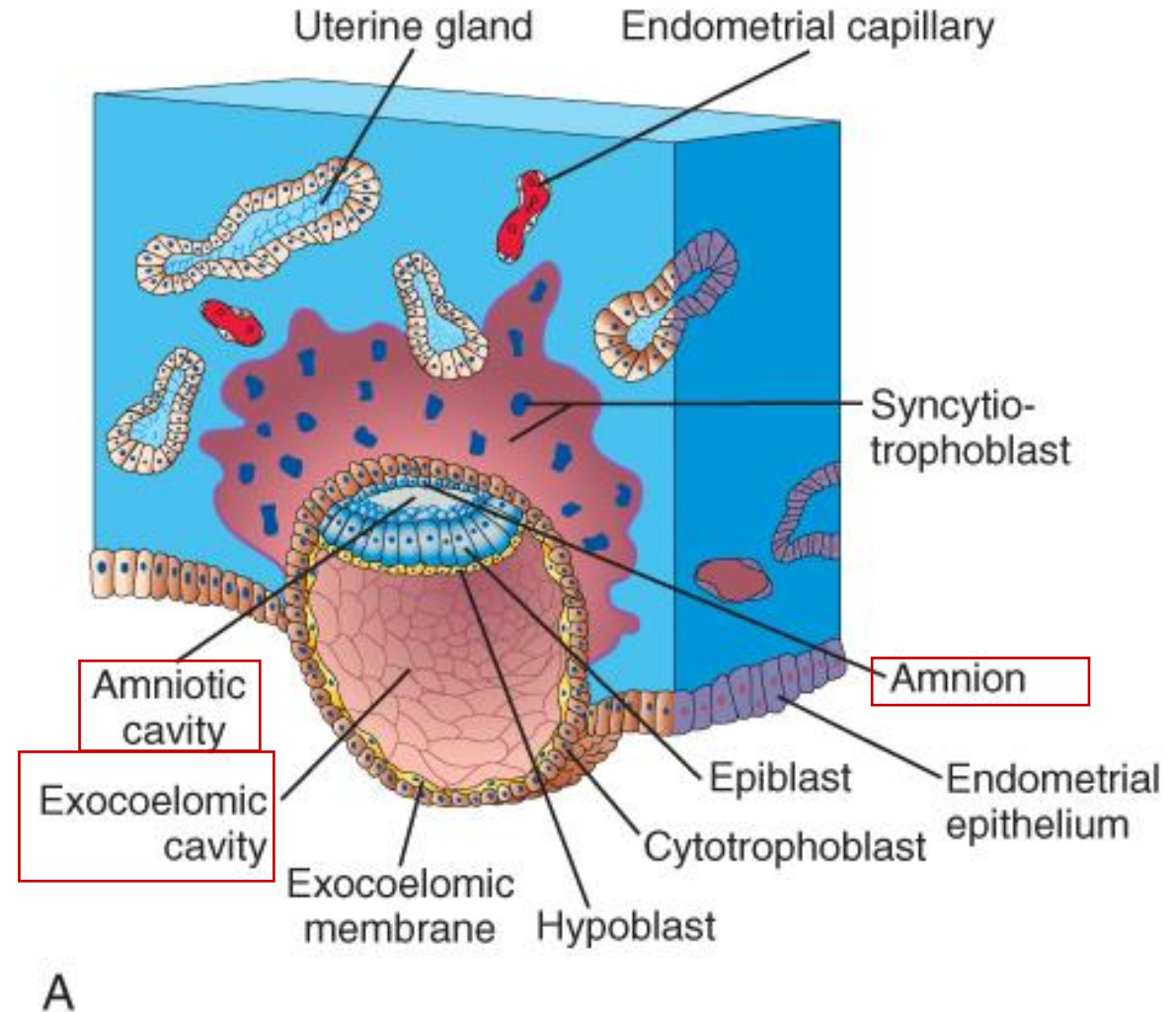
The **allantois** is an extension of the posterior wall of the yolk sac. It degenerates to form the median umbilical ligament in the adult.

- **Chorion**

The chorion is the outermost fetal membrane around the embryo which develops from an outer fold on the surface of the yolk sac.

Review Week 2 : Amnion and Amniotic Cavity (Amniotic Sac)

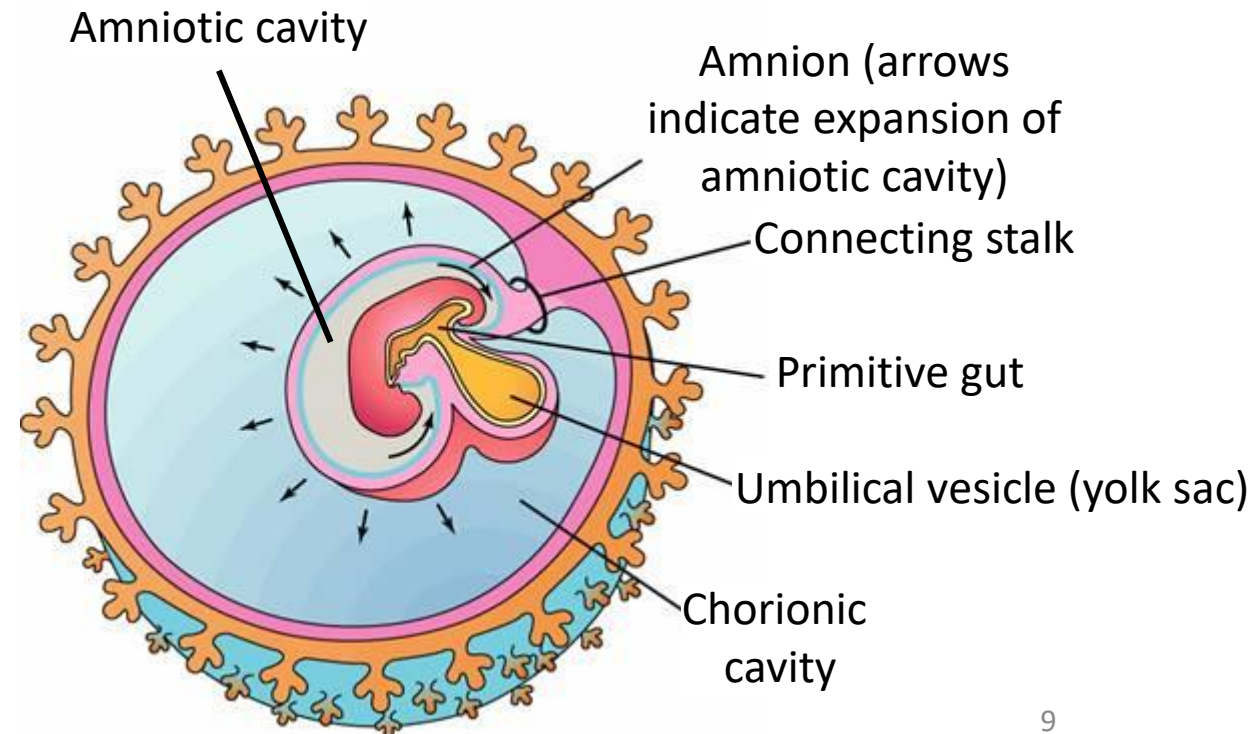
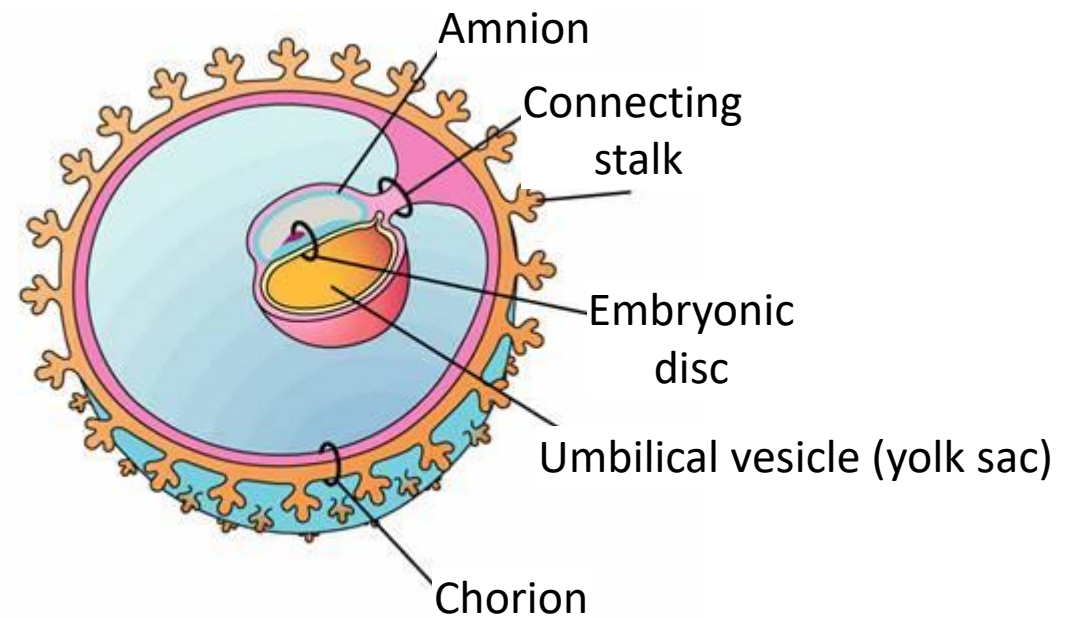
- As implantation of the blastocyst progresses, a small space appears in the embryoblast, which forms the **amniotic cavity**
- At the same time, morphologic changes occur in the **embryoblast** that result in the formation of a flat, circular bilaminar plate of cells, the **embryonic disc**, consisting of two layers
 - **Epiblast**
 - **Hypoblast**
- Soon, amniogenic (amnion-forming) cells, **amnioblasts**, separate from the epiblast and form **the amnion**, which **forms the roof of the amniotic cavity (the remaining epiblast form the floor of the cavity)**



Amniotic fluid

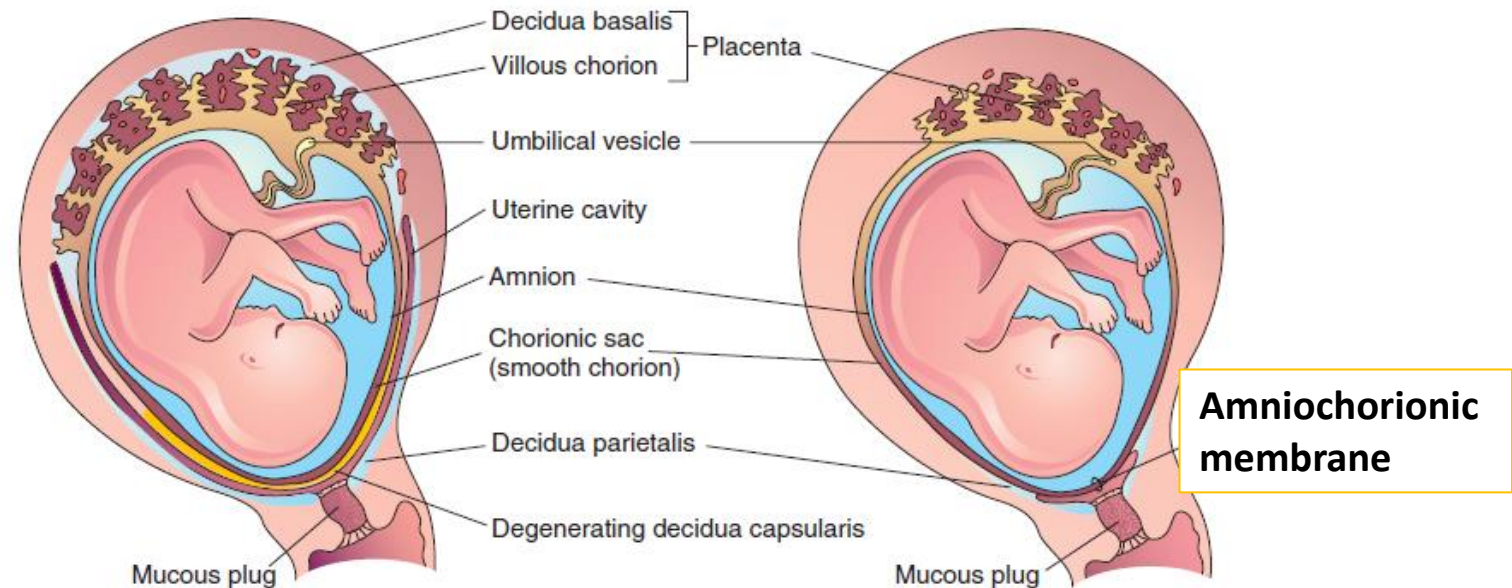
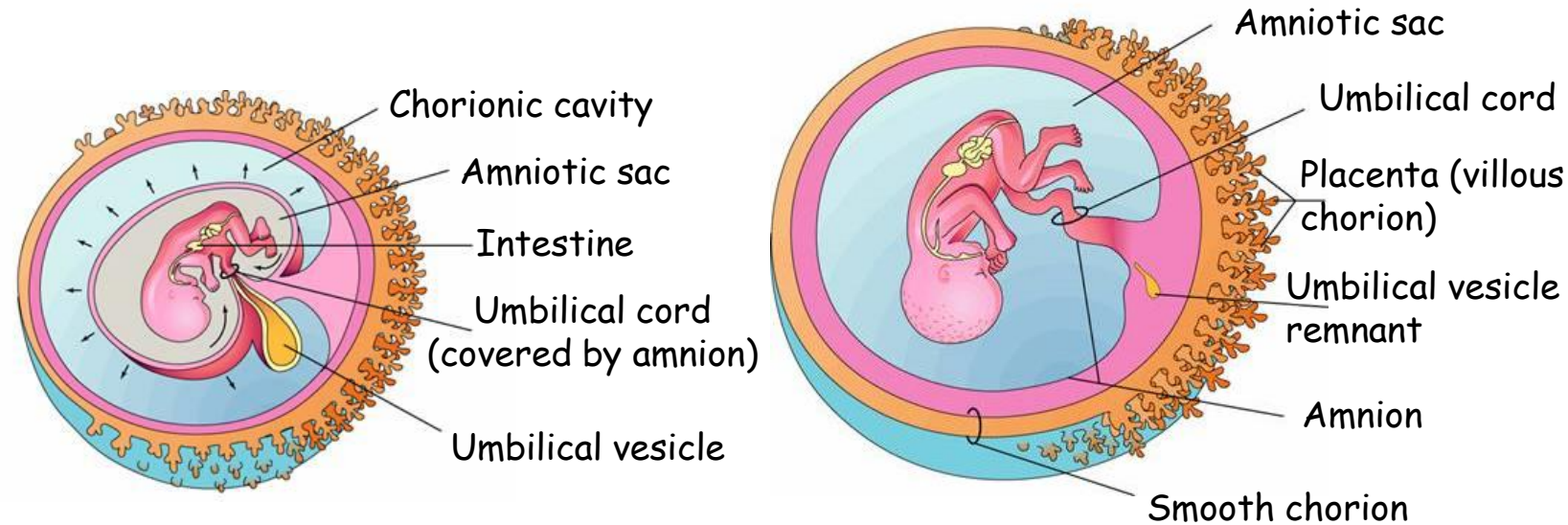
Functions:

- Shock absorber
- Maintains constant temperature
- Movements
- Symmetrical growth
- Lung development
- Antibacterial activity
- Hydrostatic wedge



Amniotic sac

- As the fetus grows, amniotic fluid volume increases
- Amniotic sac enlarges faster than the chorionic sac
- The amnion eventually fuses with chorion wall thus obliterating chorionic cavity and forming the single **amniochorionic membrane**
- Fetus now floats in amniotic fluid with cord attaching it to placenta



Disorders of Amniotic Fluid Volume

- Oligohydramnios

- Low volume

- < 400-500 ml in 3rd trimester
 - Inadequate amount

- Causes

- Preterm rupture of amniochorionic membrane
 - About 10% of pregnancies
 - Urinary system anomalies
 - Bilateral renal agenesis
 - Placental insufficiencies
 - Decreased blood circulation

- Approximately 4% of pregnancies in US

- Complications

- Lung hypoplasia
 - Facial or limb deformities
 - Due to fetal compression by the uterine wall

- Polyhydramnios

- High volume

- > 2000 ml in 3rd trimester
 - Excessive amount

- Causes

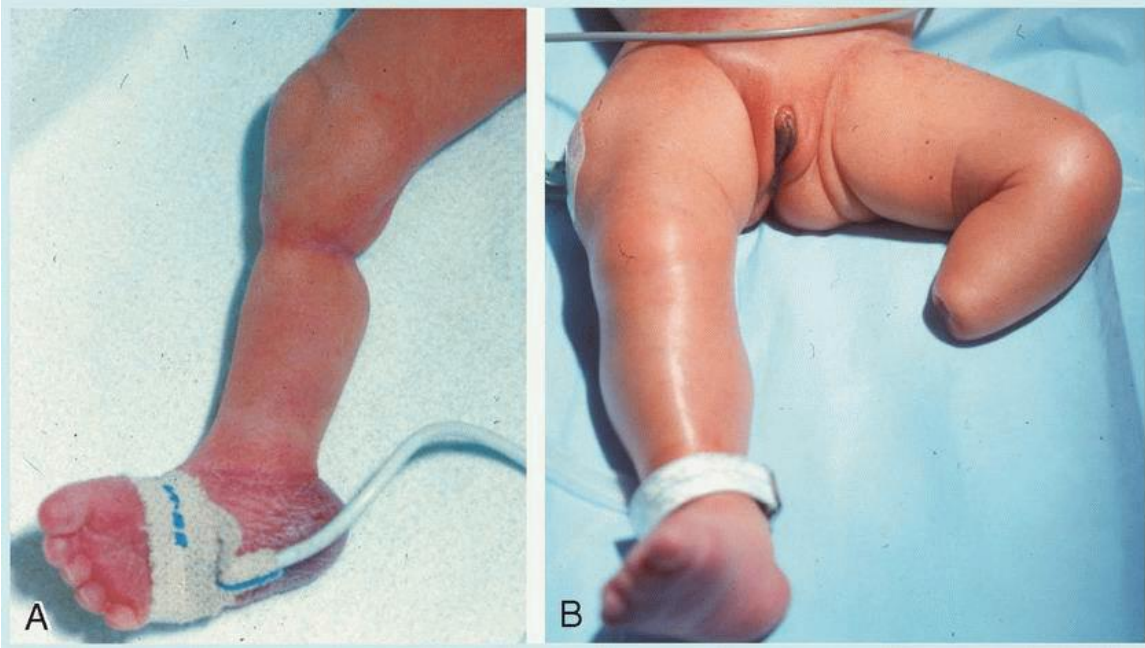
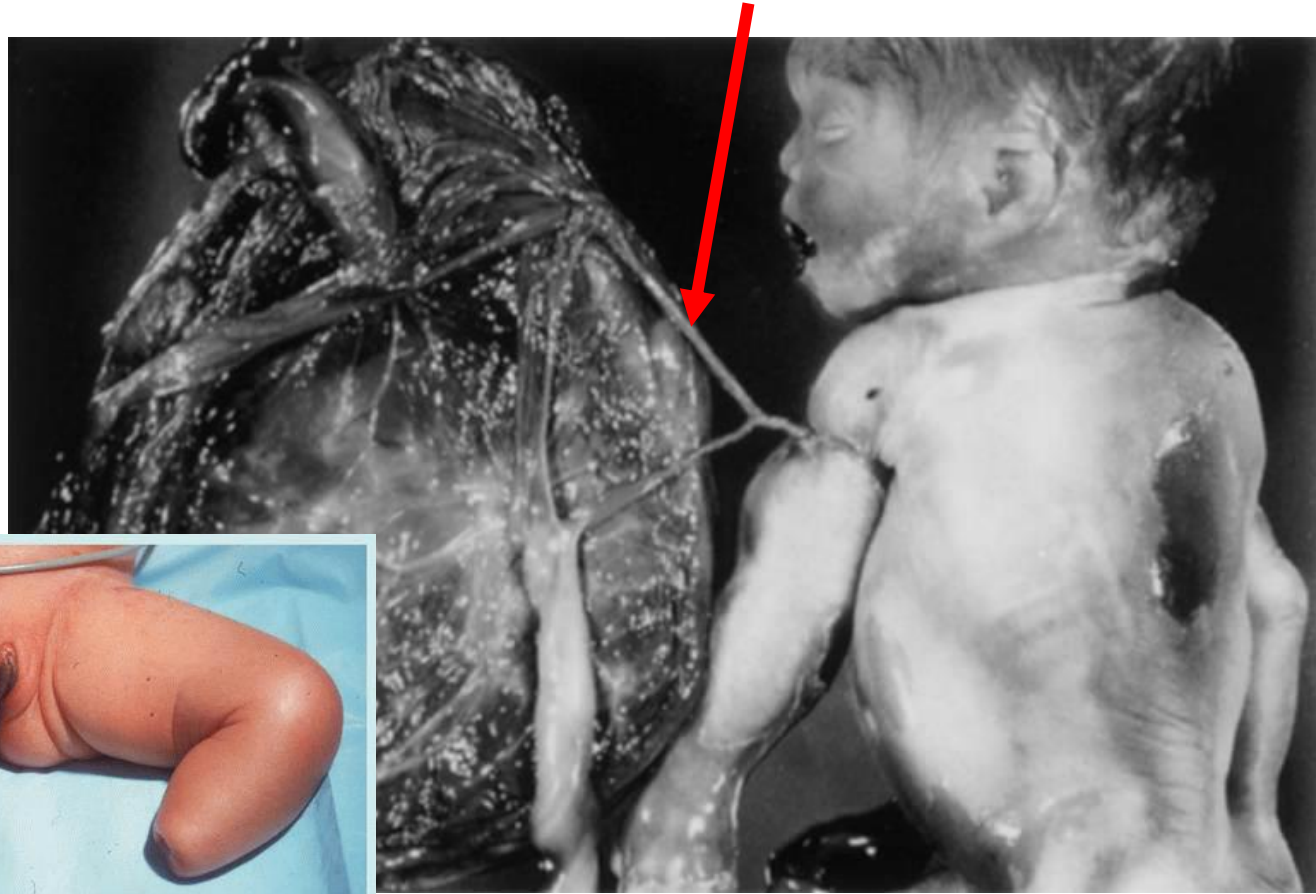
- Idiopathic (60%)
 - Maternal (20%)
 - Multiple pregnancies
 - Maternal diabetes
 - Fetal (20%)
 - Digestive system anomalies
 - » Esophageal atresia
 - CNS anomalies
 - » Meroencephaly
 - » Anencephaly

- Approximately 1% of pregnancies in US

- Complications

- Preterm labor & delivery
 - Premature rupture of amniochorionic membrane

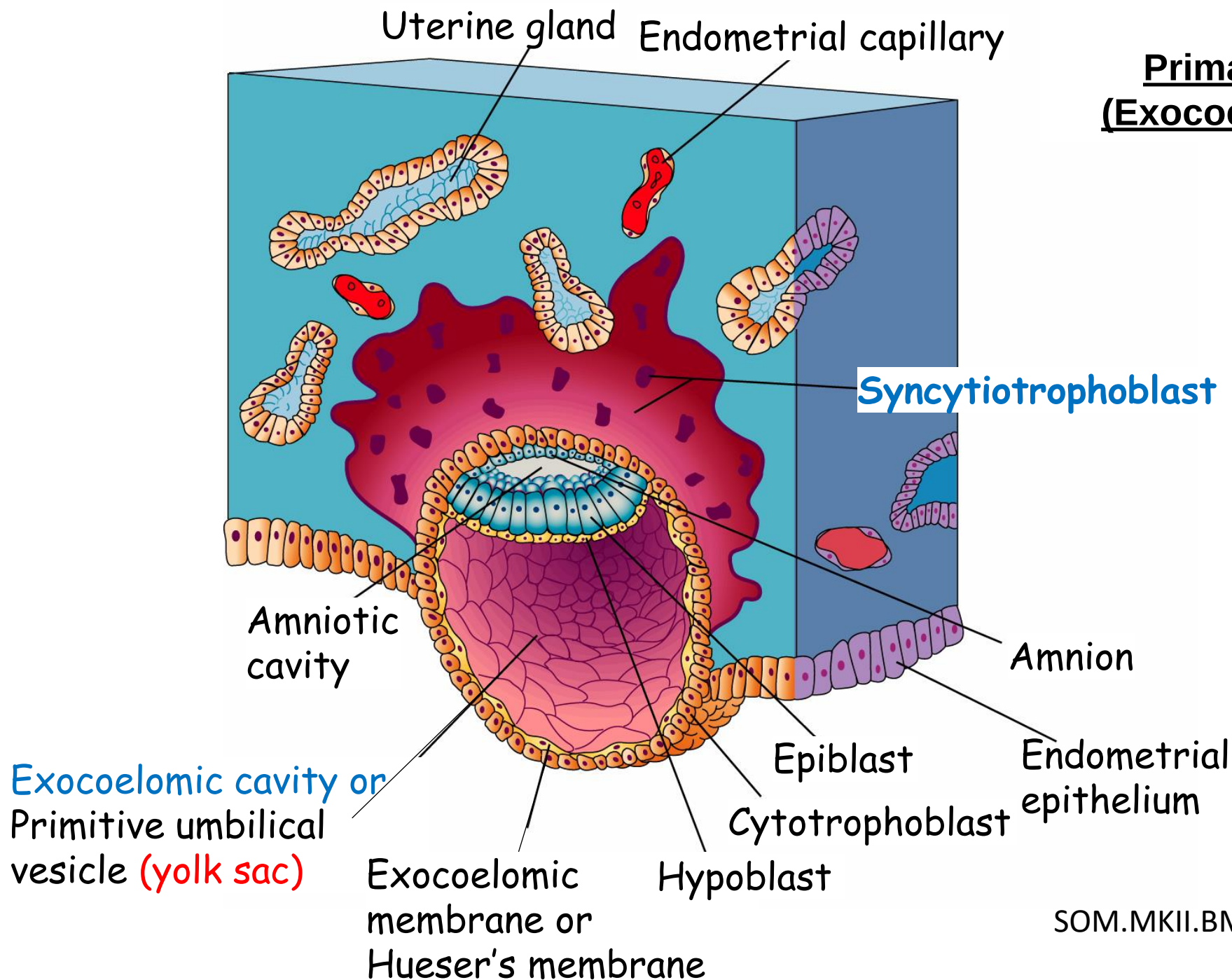
Amniotic band syndrome – amniotic bands can amputate the limbs



An 18-year-old woman comes to the physician for a follow-up examination at 32-weeks gestation. Her weight and fundal height have not increased since her last visit and the baby has not been moving as frequently as usual. Ultrasonography shows reduced amniotic fluid. Which of the following is most likely a complication of the patient's condition?

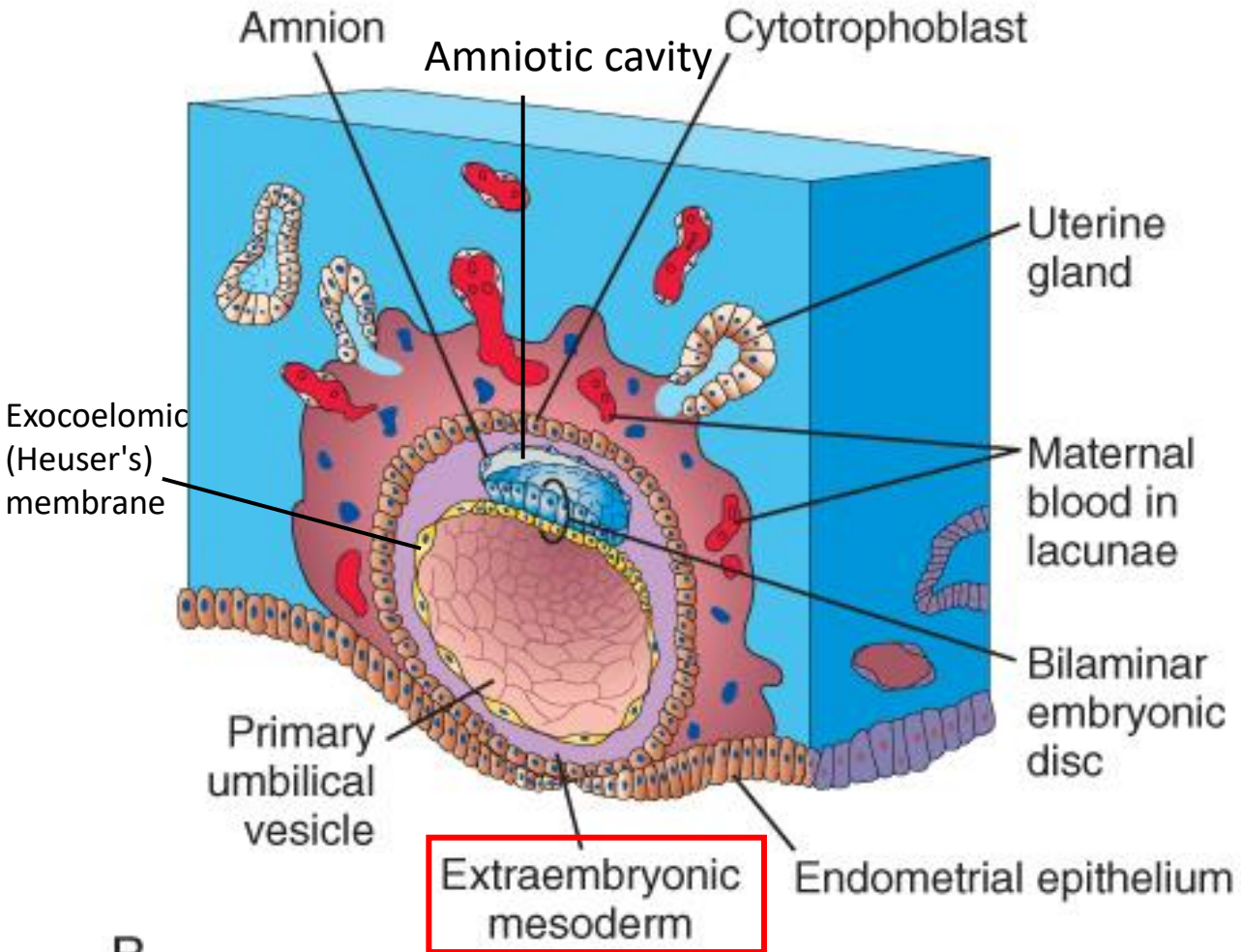
- a. Esophageal atresia
- b. Preterm labor
- c. Fetal pulmonary hypoplasia
- d. Anencephaly
- e. Renal agenesis

Review:
Primary umbilical vesicle
(Exocoelomic cavity/yolk sac)



- First site of blood formation
- Gives rise to the primordial gut
- Primordial germ cells arise in the endodermal lining of the yolk sac

Chorion



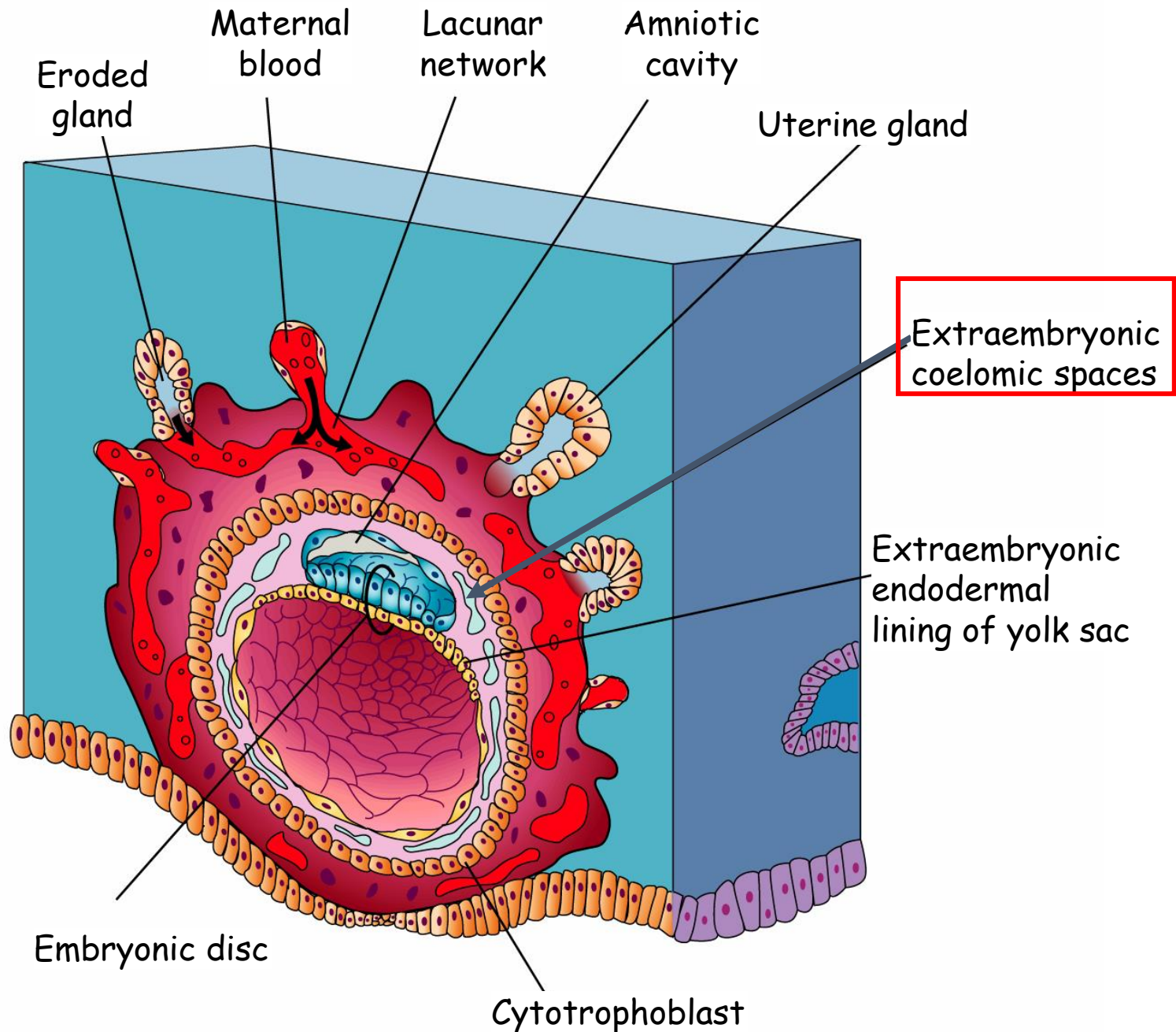
Extraembryonic (primary) mesoderm fills the space between the trophoblastic wall & and the amniotic sac and the primary umbilical vesicle. This surrounds the amnion and primary umbilical vesicle.

B

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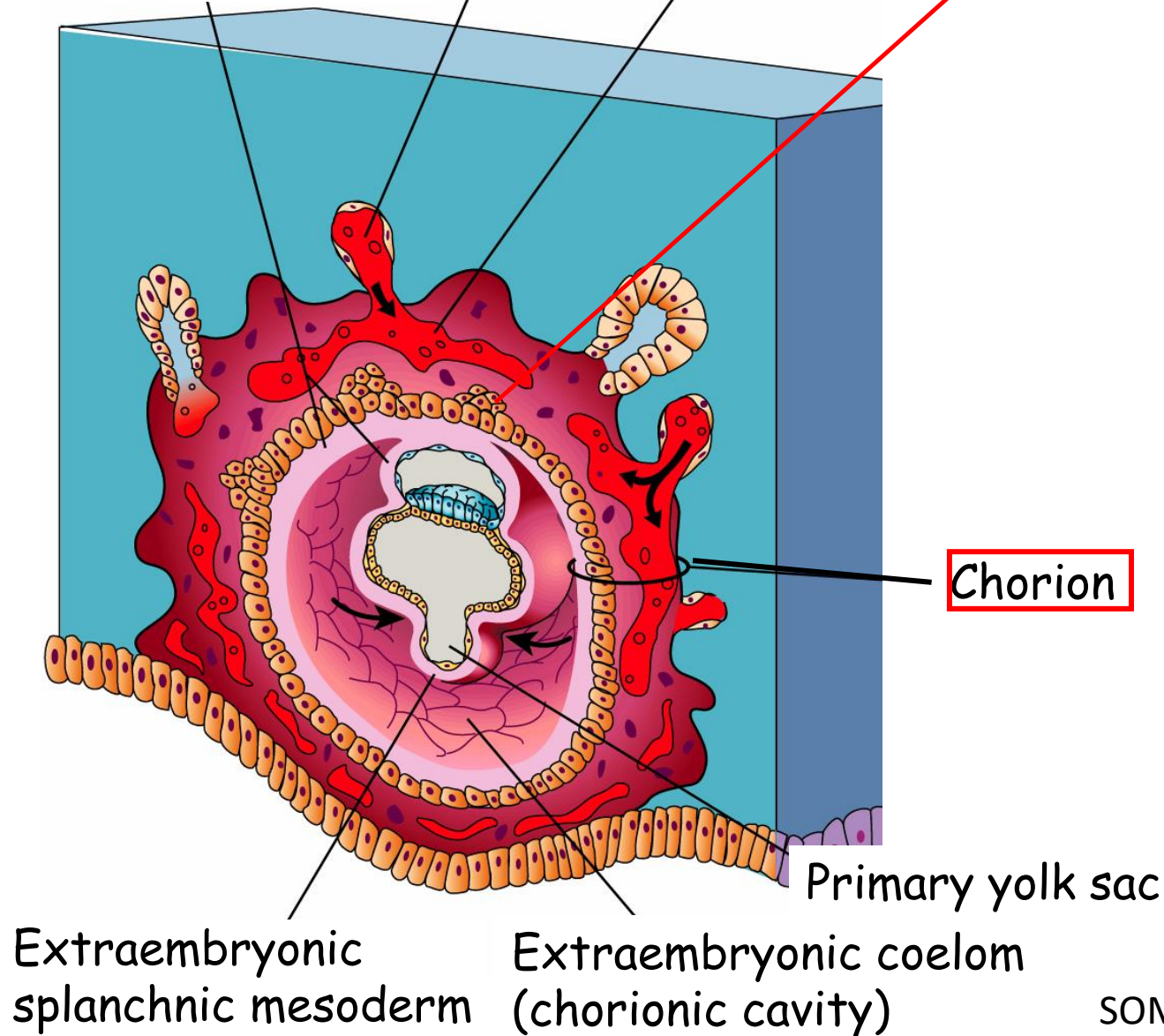
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Formation of Chorionic Cavity

- Clefts appear in the extraembryonic mesoderm called extraembryonic coelomic spaces
- Clefts fuse to form a cavity called the **extraembryonic coelom the precursor to the chorionic cavity**
- **Extraembryonic coelom** splits the extraembryonic mesoderm into two layers:
 - **extraembryonic splanchnic mesoderm → surrounds the umbilical vesicle**
 - **extraembryonic somatic mesoderm → lines trophoblast and covers amnion**

Extraembryonic somatic mesoderm Maternal sinusoid Lacunar network Primary chorionic villus



Chorion sac and wall

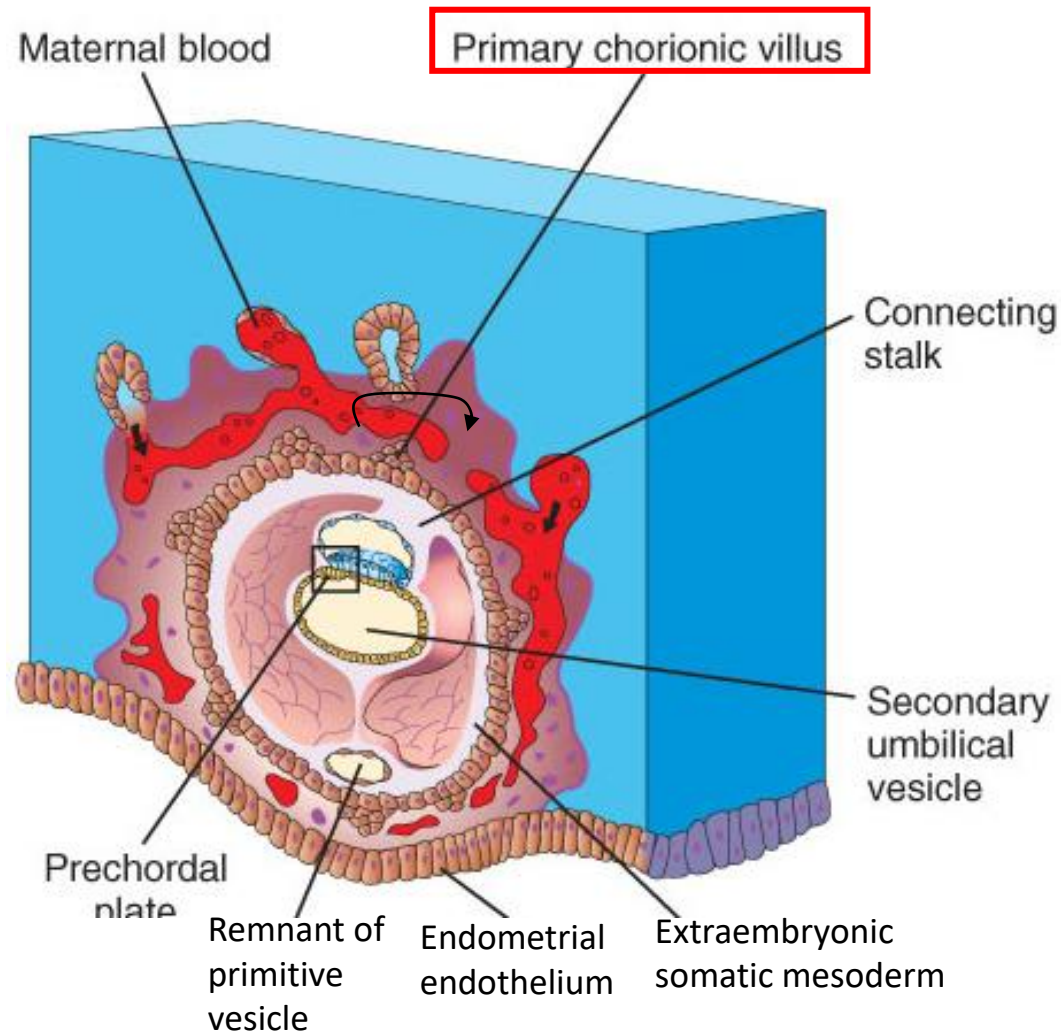
Chorion sac :

- Chorion wall
- Chorionic cavity

Chorion wall is made of:

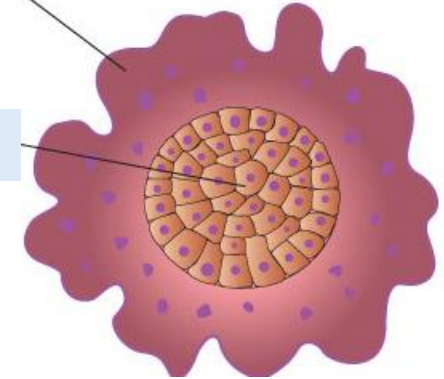
- Syncytiotrophoblast
- Cytotrophoblast
- Somatic layer of extraembryonic mesoderm

Development of chorionic villi



Syncytiotrophoblast

Cytotrophoblast



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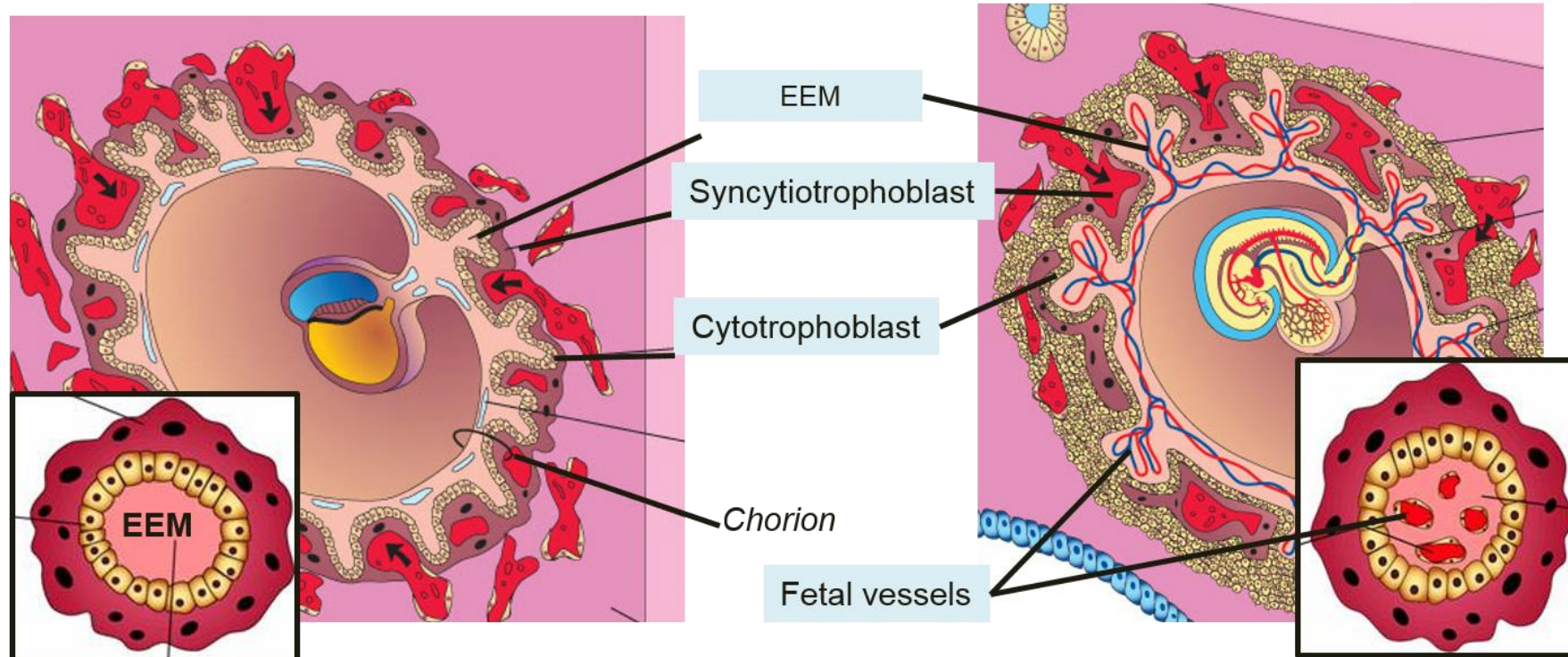
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- Chorion wall presents finger-like projections called villi
- There is local proliferation of cytotrophoblast cells into the syncytiotrophoblast → **primary chorionic villi**

Development of chorionic villi

Secondary chorionic villus

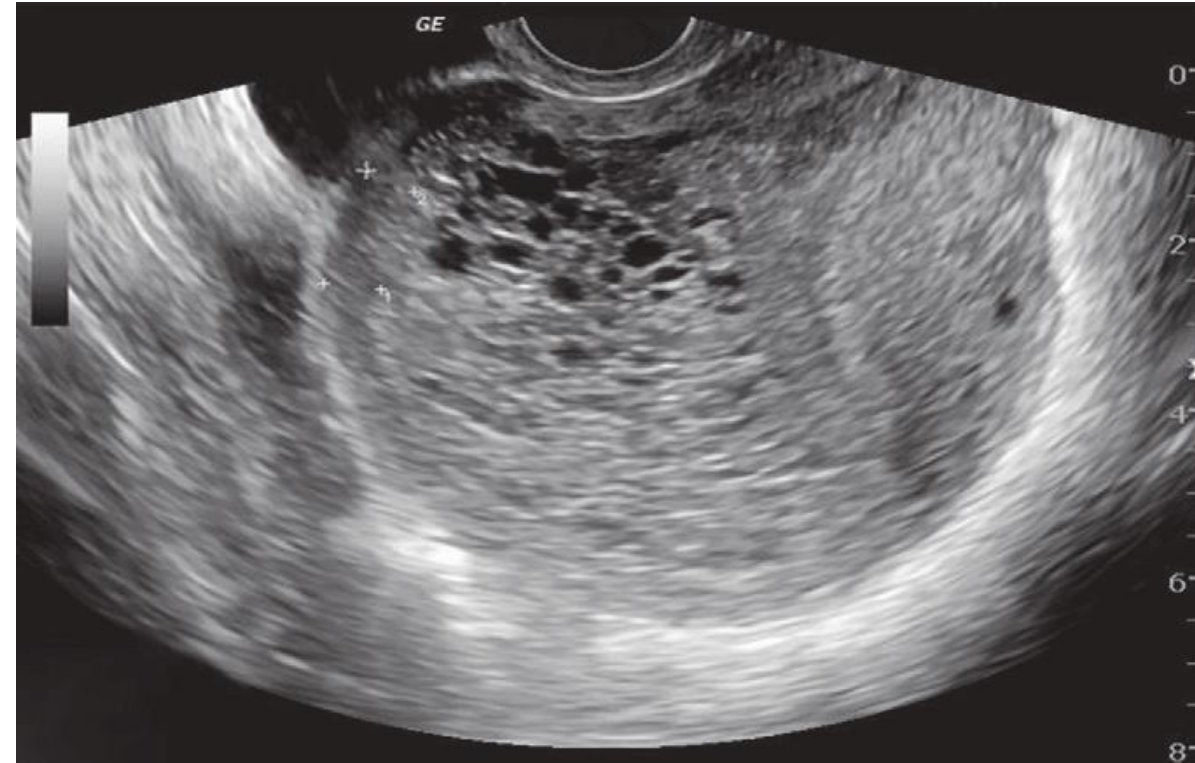
Tertiary chorionic villus



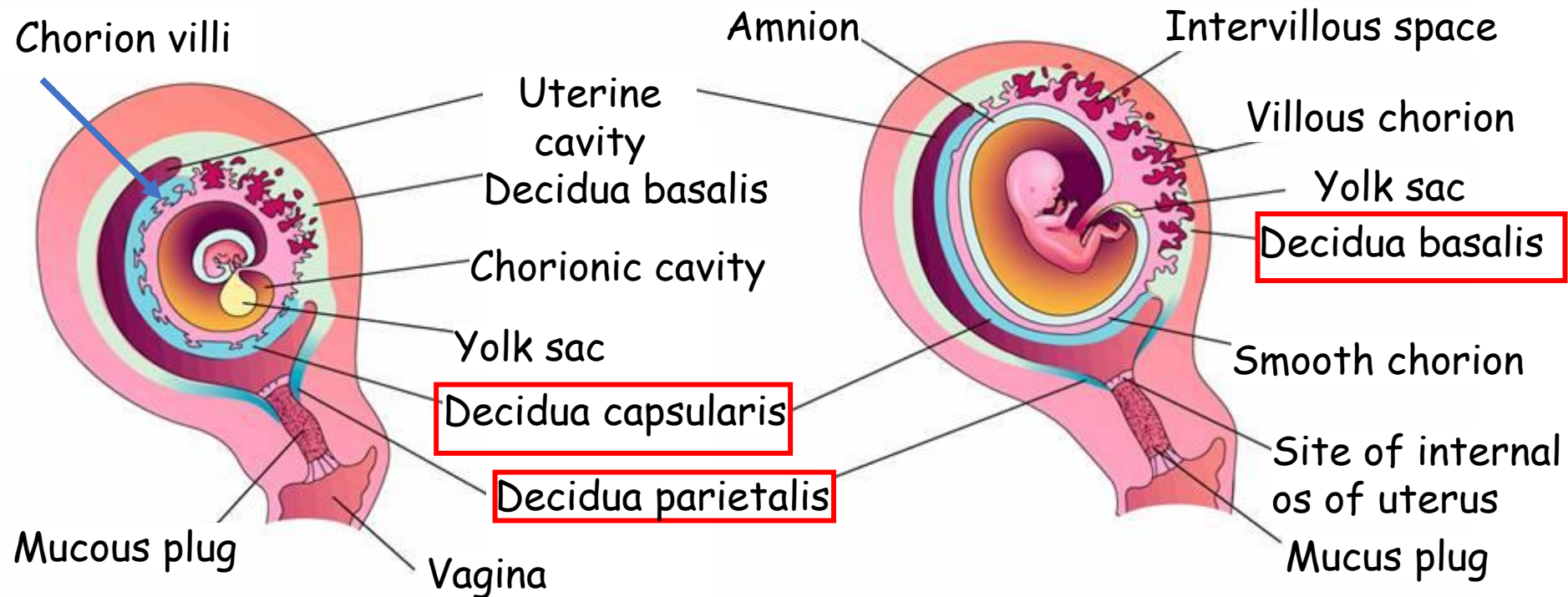
- Penetration of extraembryonic mesoderm into the core of the primary chorionic villi → **secondary chorionic villi**
- Blood vessels develop in the extraembryonic mesoderm core of the 2nd villi → **tertiary chorionic villi**

Hydatiform Mole

- Sometimes the embryo dies and the chorionic villi do not complete their development; that is, they do not become vascularized to form tertiary villi.
- These degenerating villi may form cystic swellings, called hydatidiform moles. These moles exhibit variable degrees of trophoblastic proliferation and produce excessive amounts of human chorionic gonadotropin.
- In about 5% of such cases, these moles develop into malignant trophoblastic lesions, called choriocarcinomas.



The Decidua (endometrium of pregnant uterus)



- **Decidua basalis** is the endometrium deep to and in contact with villous chorion
 - **Decidua capsularis** is the endometrium related to the smooth chorion & separates it from the uterine cavity. It overlies the conceptus.
 - **Decidua parietalis** refers to the rest of endometrium not directly related to the chorion sac
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Placenta

General description:

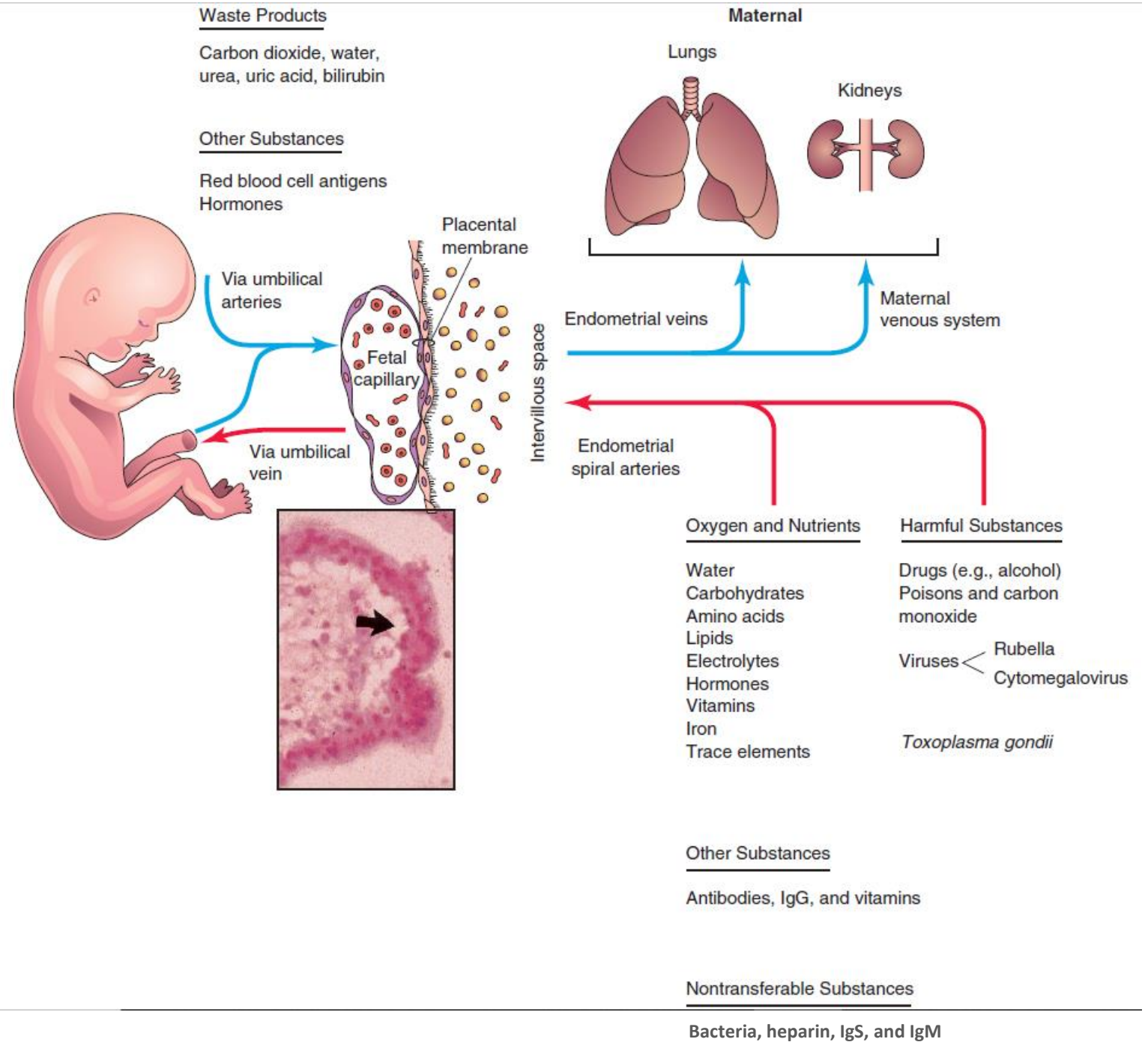
- Discoid in shape
- 20 cm in diameter
- 3 cm thick
- ½ kg in weight

Formation:

- Maternal component – formed from the decidua basalis
- Fetal component – formed from the villous chorion. Attached to the maternal part of placenta by a cytotrophoblastic shell
 - Cotyledons – placental septa (wedge shaped areas of decidua) project toward the chorionic plate (“roof” of fetal placenta) and divide the fetal parts of the placenta into irregular convex areas

Functions of the Placenta

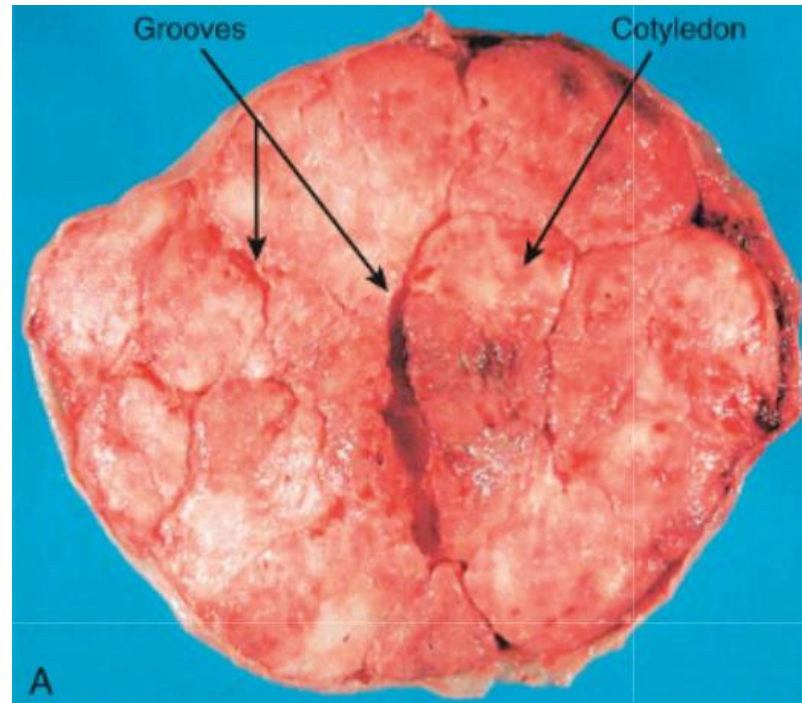
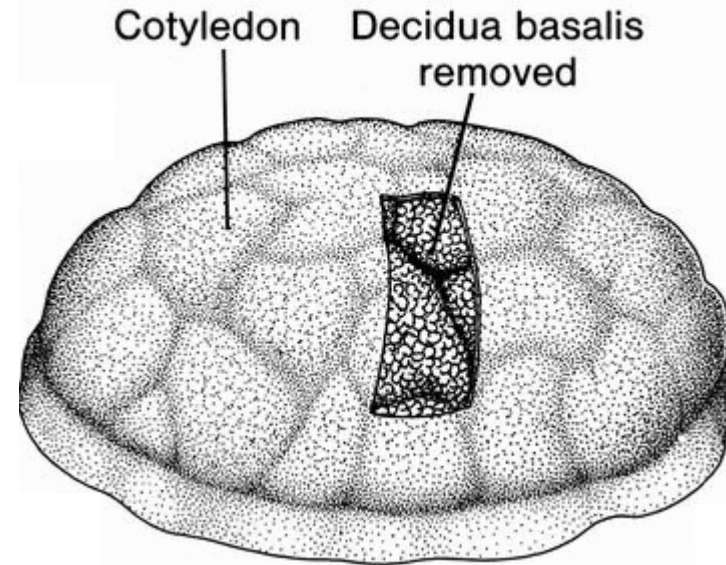
- Protection
- Nutrition
- Respiration
- Excretion
- Hormone production



Placenta after birth

Maternal surface:

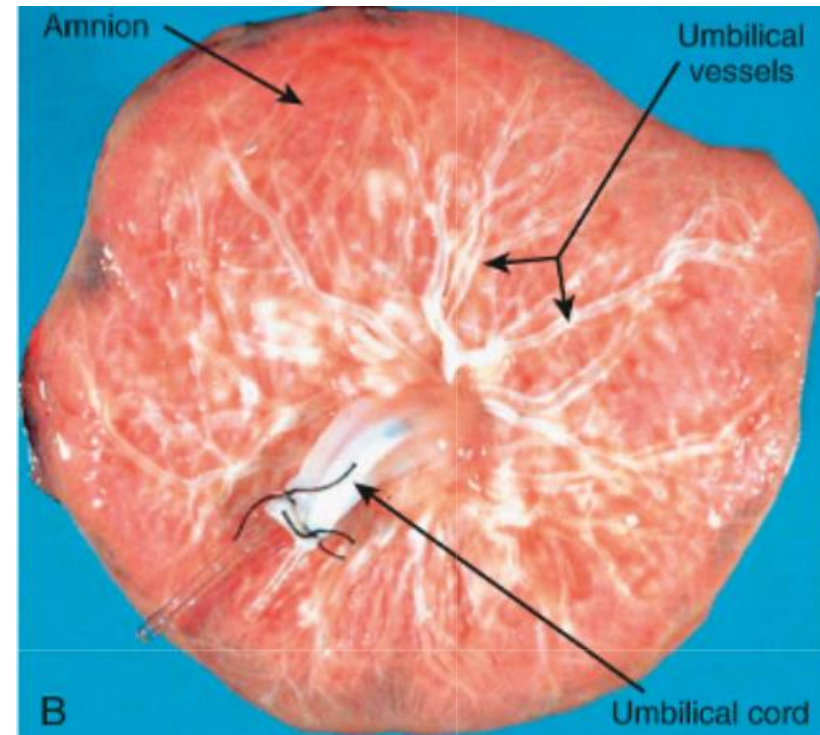
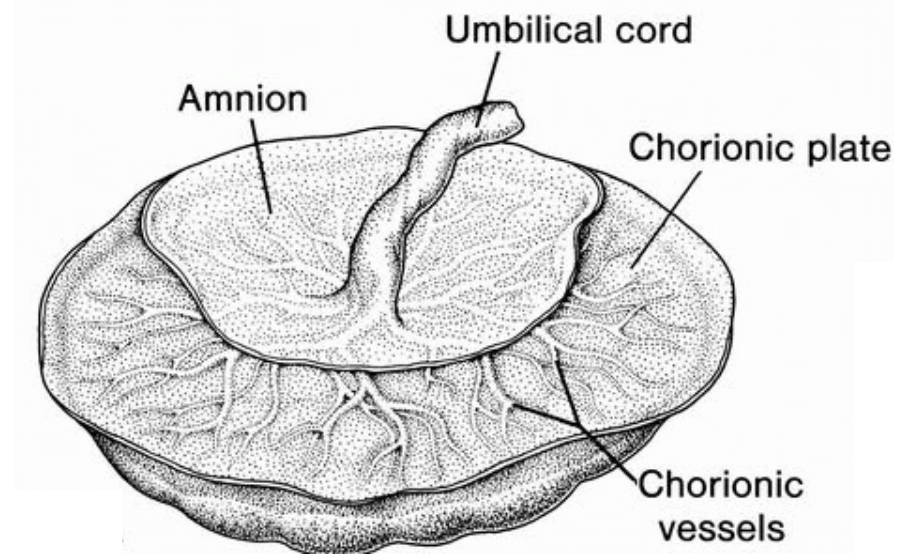
- rough shows fetal cotyledons demarcated by grooves (formerly occupied by placental septae)



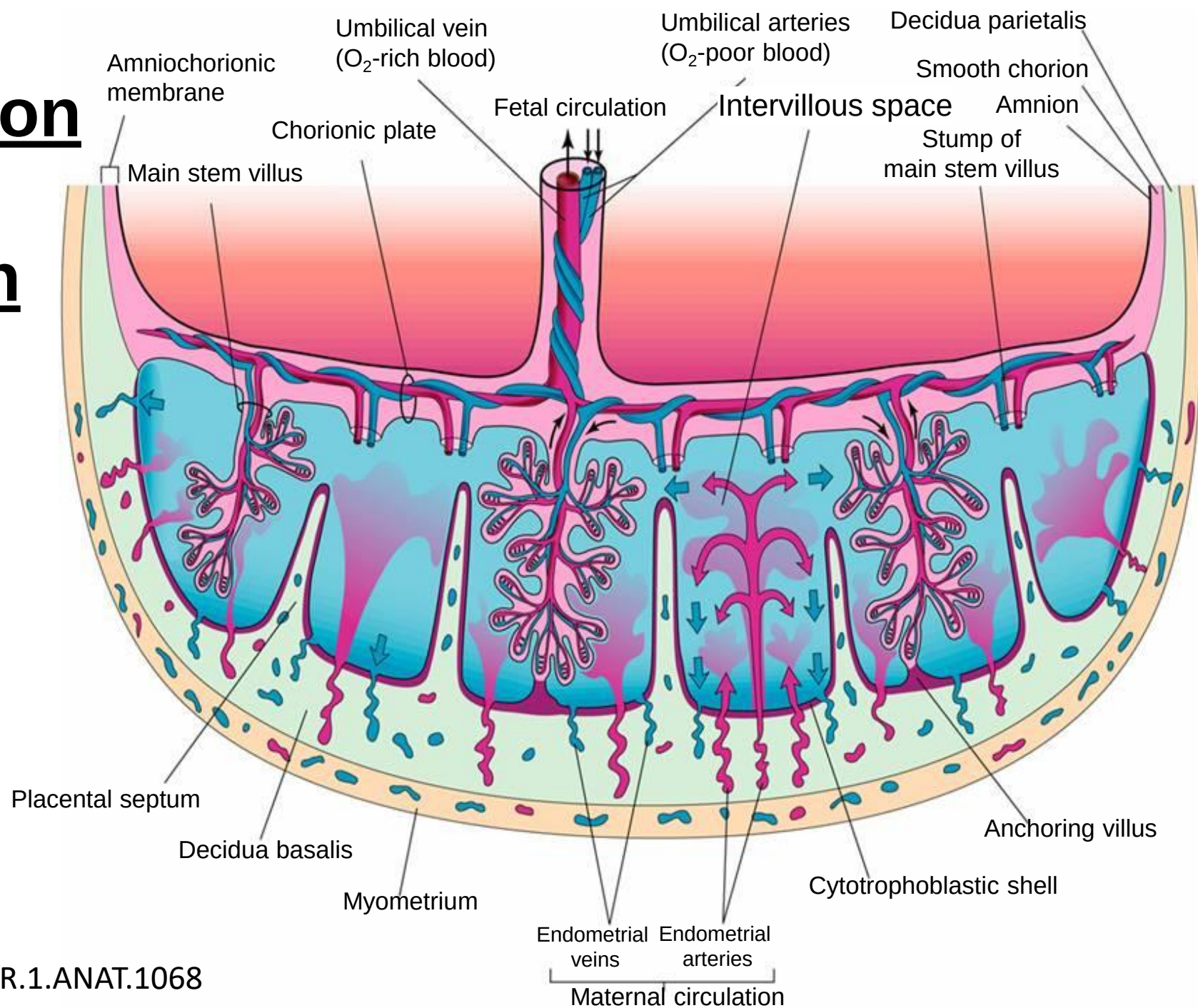
Placenta after birth

Fetal surface:

- Smooth, covered with the transparent amniotic membrane
- Umbilical cord is attached to this surface
- Umbilical vessels - Branches of the vessels ramify on this surface deep to the amnion



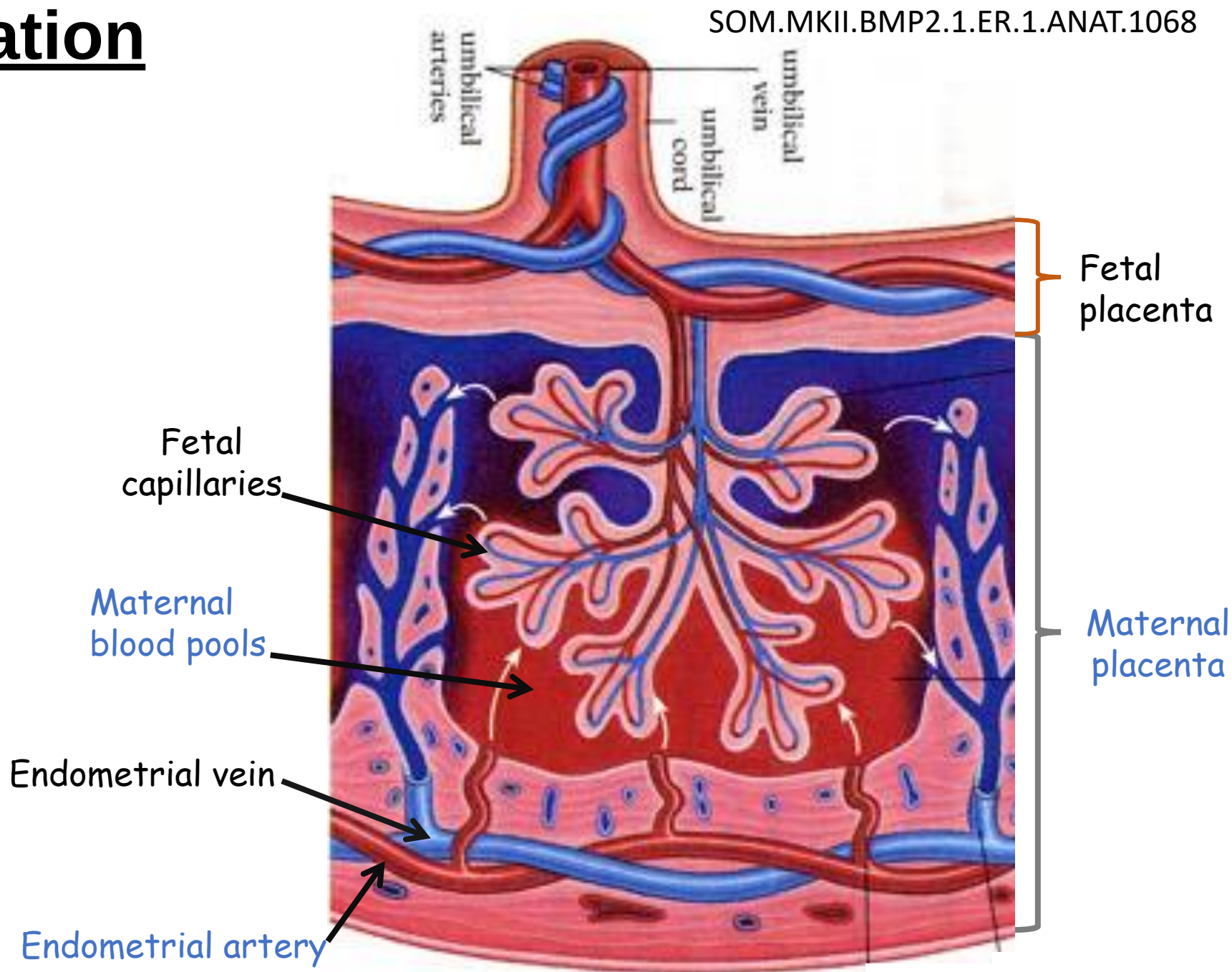
Placental organization and circulation



Placental circulation

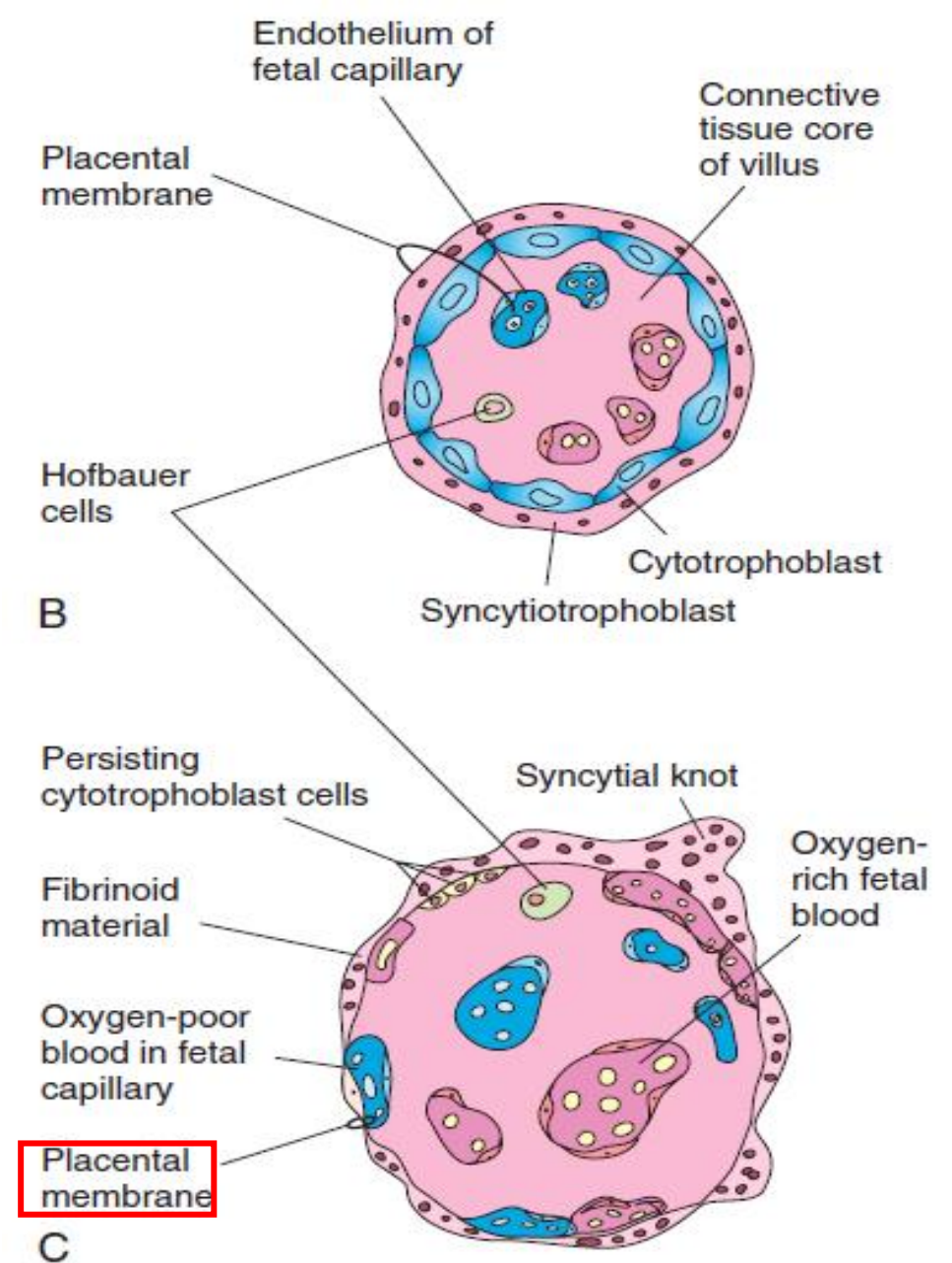
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- The intervillous spaces are filled with maternal blood from the spiral arteries (branches of uterine arteries)
- Tertiary villi float in the blood in the intervillous space
- Gaseous exchange takes place between the maternal blood & blood in fetal capillaries
- **No intermingling of fetal and maternal blood**



Placental membrane

- The placental membrane consist of extra fetal tissue separating the maternal and fetal blood.
- Initially consists of 4 layers:
 - **Syncytiotrophoblast, cytotrophoblast, connective tissue of the villi, endothelium of fetal capillaries.**
- After 20 weeks, cytotrophoblast disappear leaving the membrane consisting of 3 layers
- In some areas, there is thinning of the membrane where the syncytiotrophoblast comes in direct contact with the fetal endothelium to form a **vasculosyncytial placental membrane**
- The membrane becomes thinner as pregnancy progresses



Sections through tertiary villi at 10 weeks (B) and at full term (C).

Hemolytic disease of the newborn

- Some fetal blood may pass to the maternal blood through microscopic breaks in the placental membrane.
- If the fetus is Rh-positive and the mother is Rh-negative, the fetal cells may stimulate the formation of anti-Rh antibody by the mother's immune system.
- This can cause hemolysis of fetal Rh-positive blood cells and anemia in the fetus.
- Fetus may die unless delivered early or given transfusions of packed Rh-negative blood cells.
- Relatively uncommon.

Umbilical Cord



Which type of connective tissue is in the umbilical cord?

- Short cords can be ruptured
- Long cord can twist around the neck
- Long cords can form knots
- Aberrant number of umbilical vessels

Placental abnormalities

- **Placenta accreta**

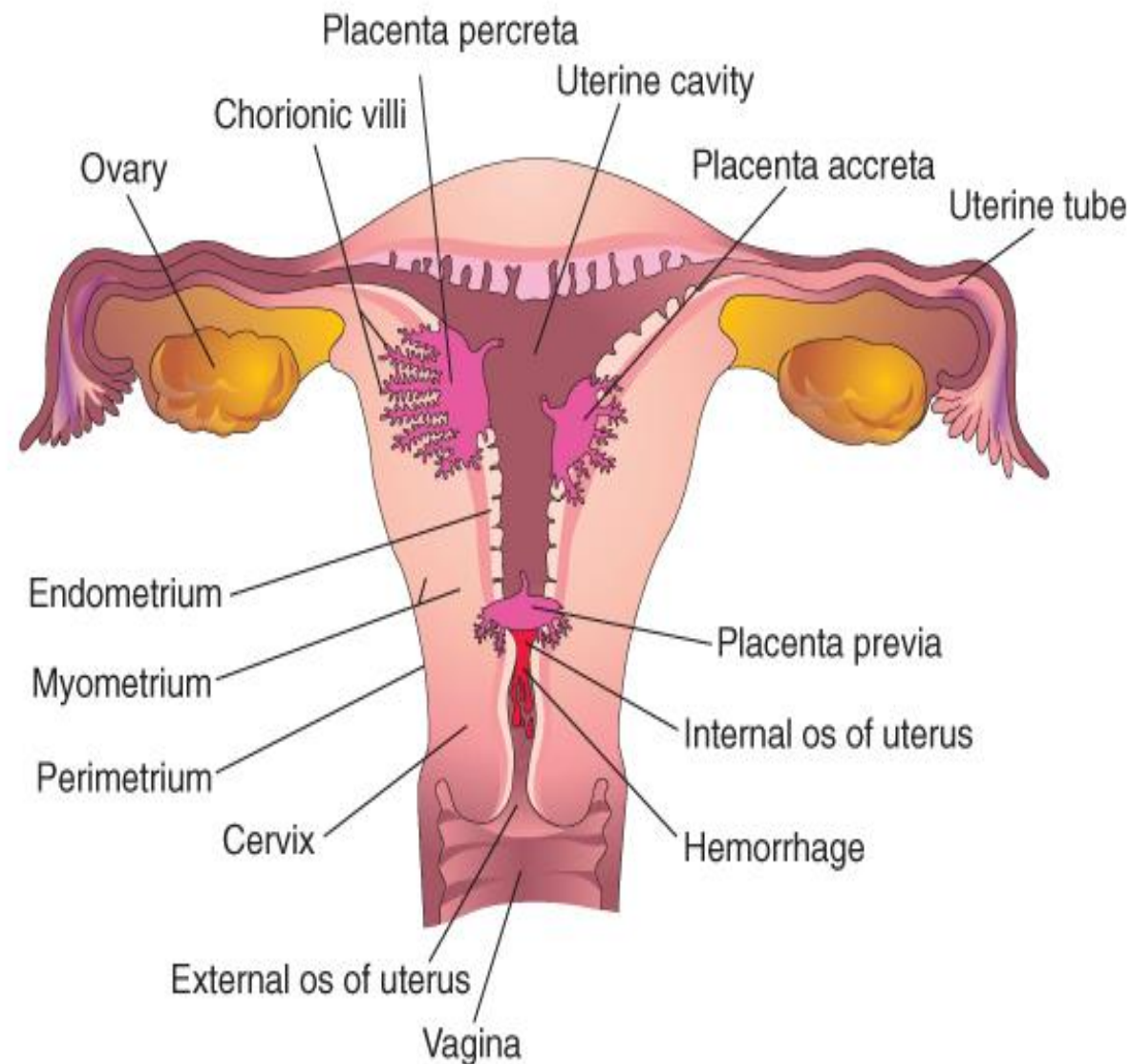
- Abnormal adherence of chorionic villi **to myometrium** of the uterine wall.
- **Placenta increta** – chorionic villi invade deeply into the myometrium

Placenta percreta

- Chorionic villi penetrates the entire uterine wall **to and/or through the perimetrium.**
- 3rd trimester bleeding.

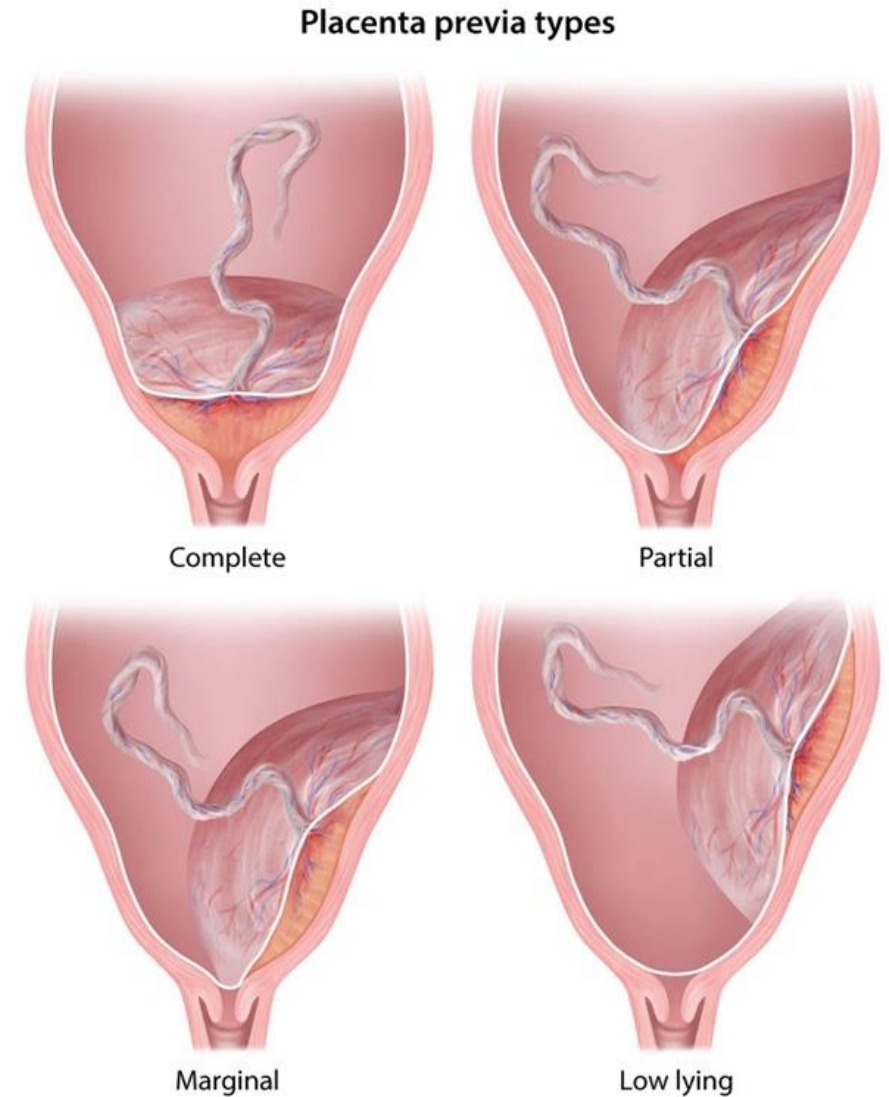
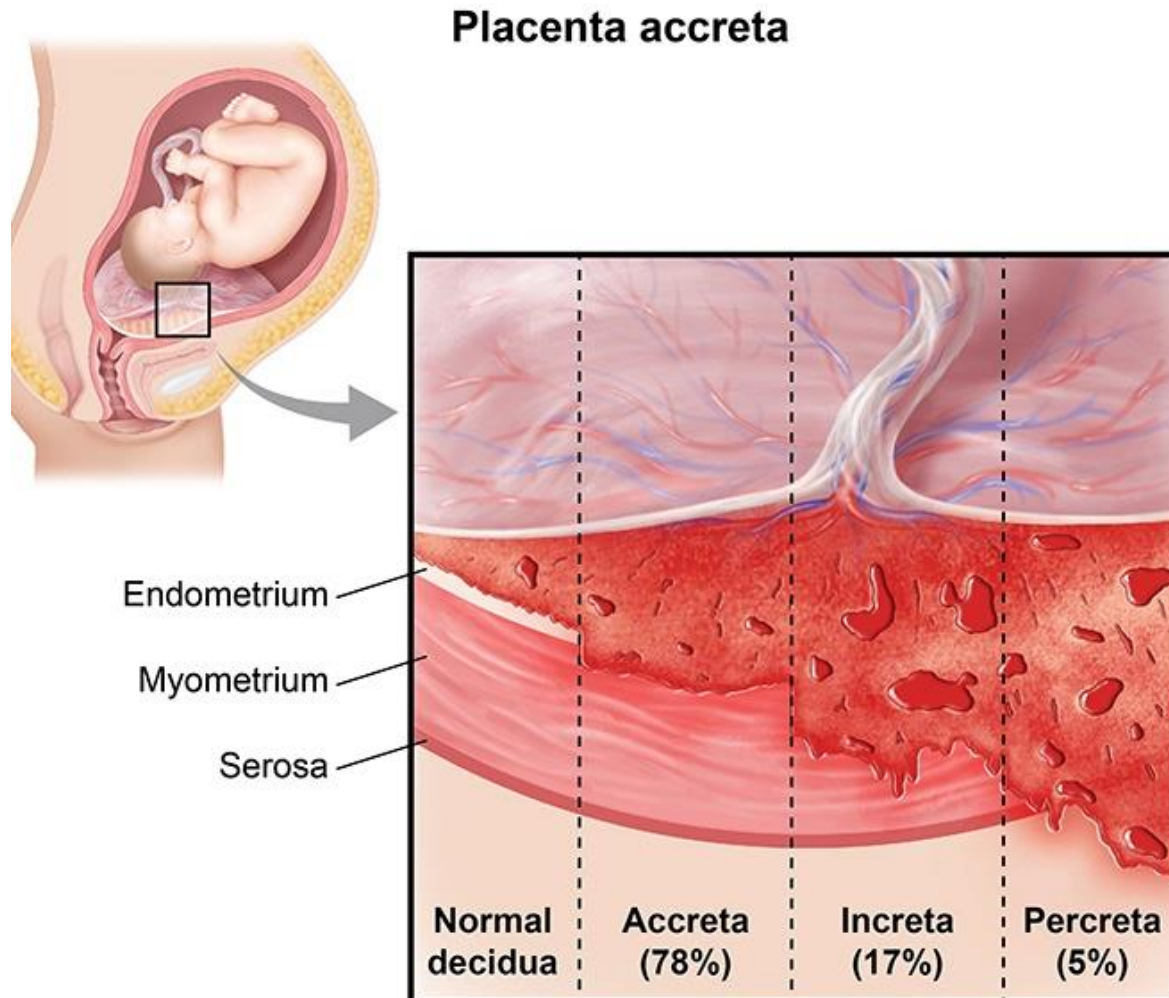
- **Placenta previa**

- **Implantation close to or overlying the internal os**
- Placenta bridges the cervical opening.
- Severe bleeding in later part of pregnancy



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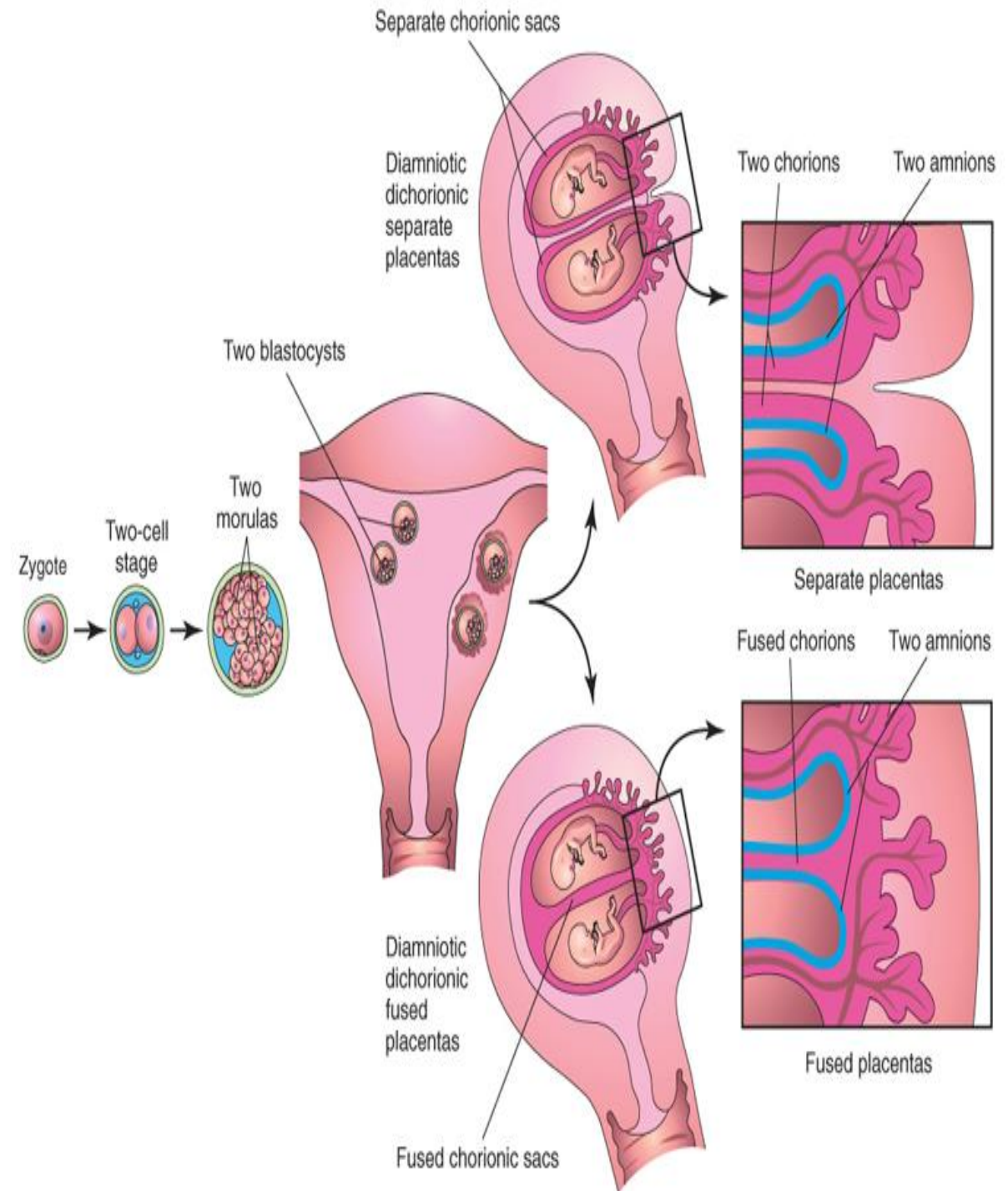
Placental abnormalities



Twin pregnancies

• Dizygotic twins

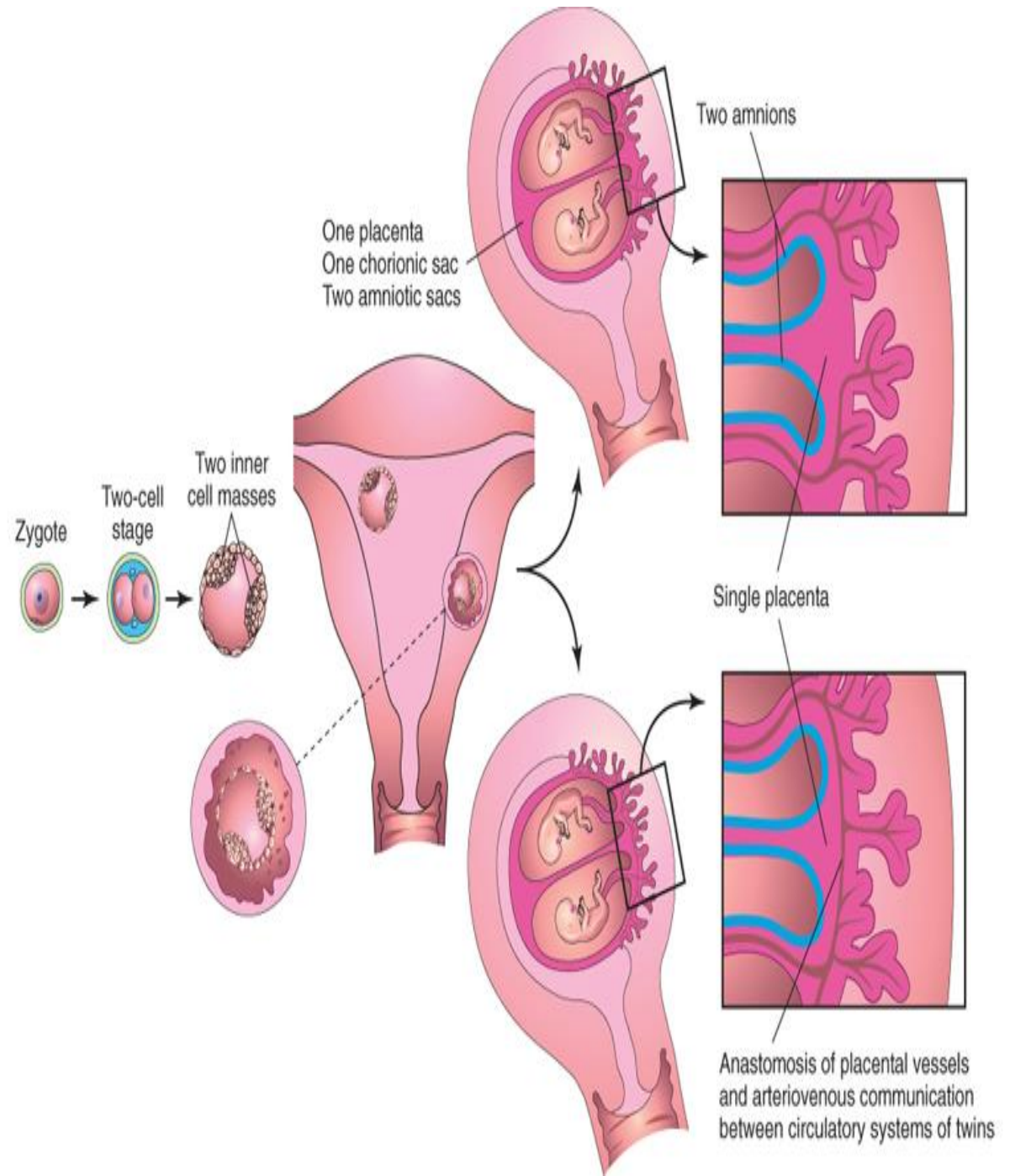
- Most common type.
- Origin from 2 zygotes.
- Fertilisation of 2 oocytes by 2 sperms.
- 2 amnions and 2 chorions.
- Placentas may be fused
- May be of the same sex or different sexes.



Twin pregnancies

- **Monozygotic twins**

- Origin from a single zygote
- Genetically identical and same sex
- Division of embryoblast into two embryonic primordia (occurs at the blastocyst stage)
- 2 amniotic sacs
- 1 chorionic sac
- 1 placenta



Abnormalities of Twinning

- **Twin transfusion syndrome**

- Preferential shunting of arterial blood in one twin to venous blood in the other.
- Donor twin (small and anemic)
- Recipient twin (large and polycythemic)

- **Conjoint twins**

- Incomplete division of the embryonic disc or fusion of adjacent embryonic discs



Twin transfusion syndrome

A 36-year-old pregnant woman comes to the physician for a routine ultrasound at 20 weeks gestation. Physical examination shows active fetal movements. Ultrasound shows the placenta has invaded deeply into the myometrium and has penetrated the perimetrium. Which of the following is the most likely diagnosis?

- a. Placenta previa
- b. Placenta accreta
- c. Placenta percreta
- d. Placenta increta
- e. Hydatiform mole

The maternal component of the placenta

1. Consists of the decidua basalis, which is derived from the endometrium of the uterus located between the blastocyst and the myometrium.
2. The decidua basalis and decidua parietalis (which includes all portions of the endometrium other than the site of implantation) are shed as part of the afterbirth.
3. The decidua capsularis, the portion of endometrium that covers the blastocyst and separates it from the uterine cavity, becomes attenuated and degenerates at week 22 of development because of a reduced blood supply.
4. The maternal surface of the placenta is characterized by 8–10 compartments called cotyledons (imparting a cobblestone appearance), which are separated by decidual (placental) septa.
5. The maternal surface is dark red in color and oozes blood due to torn maternal blood vessels.

The fetal component of the placenta

1. Consists of tertiary chorionic villi derived from both the trophoblast and extraembryonic mesoderm, which collectively become known as the villous chorion.
2. The villous chorion develops most prolifically at the site of the decidua basalis. The villous chorion is in contrast to an area of no villus development known as the smooth chorion (which is related to the decidua capsularis).
3. The fetal surface of the placenta is characterized by the well-vascularized chorionic plate containing the chorionic (fetal) blood vessels.
4. The fetal surface has a smooth, shiny, light-blue or blue-pink appearance (because the amnion covers the fetal surface), and 5–8 large chorionic (fetal) blood vessels should be appear.